

Total No. of Questions : 4]

SEAT No. :

PA-10222

[Total No. of Pages : 2

[6010]-98

B.E (Mechanical Engineering) (Insem)

QUALITY AND RELIABILITY ENGINEERING

(2019 Pattern) (Semester - VIII) (402050A) (Elective - V)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4.
- 2) Figures to the right indicate full marks.
- 3) Use of electronic calculator is allowed.
- 3) Assume suitable data if necessary.

Q1) a) Explain Deming's 14 quality principles. [8]

b) What is "5S System"? Explain 5S system methodology with example.

[7]

OR

Q2) a) Explain the term & need in detail of "Quality Circle" and it's benefits. [8]

b) Elaborate term - ISO 14000 quality standard.

[7]

Q3) a) Differentiate between control chart for Variable and Attribute data. [5]

b) Table given below shows the number of defectives found in inspection of 10 lots of 100 items each. [10]

- i) Determine the control limits for P chart and state whether the process is in control.
- ii) If the point which goes outside the control limit is analyzed and eliminated, What will be the value of new control and revised fraction defectives.

Lot no.	1	2	3	4	5	6	7	8	9	10
No. of Defectives	6	3	1	4	3	0	11	5	2	3

OR

P.T.O.

Q4) a) Define term "Sampling Inspection". Explain OC curve & its Characteristics. [5]

b) In automobile filling process, 500 ml. Of certain liquid was to be filled in bags. The permissible variation is ± 5 ml. For investigating the process capability, 5 bags were taken at random from each batch for 10 successive batches and results were plotted as follows : [10]

Batch	1	2	3	4	5	6	7	8	9	10
Mean (gm)	501	498	500	503	501	500	497	502	503	496
$\bar{X}$										
Range	3	4	2	4	3	5	4	2	6	4

Establish control chart limits for $\bar{X}$ and R charts. Plot the charts and interpret the meaning.

Take $A_2 = 0.58$, $D_3 = 0$, $D_4 = 2.11$. Will process be able to meet the specifications?

